Ultimate release counts from the killed generating function

Abstract

For the two-type birth–death–conversion process with type-specific absorbing catastrophe, let $(X_{\mathrm{rel}}, Y_{\mathrm{rel}})$ be the intracellular counts released at catastrophe, with the convention $(X_{\mathrm{rel}}, Y_{\mathrm{rel}}) = (0,0)$ when catastrophe never occurs. We record their joint law, joint probability generating function, and marginals in terms of the killed (defective) generating functions $\mathcal{F}_i$ and the catastrophe-flux generators $\mathcal{J}_i$ from the catastrophe-aware PGF appendix. The construction is exact for any non-negative type-specific rates; in the generic regime $\delta_1, \delta_2 > 0$ the atom at the origin is ordinary extinction before absorption.

1 Setup and definition

We work throughout with the continuous-time two-type birth–death–conversion process of the main paper: type 1 replicates at rate λ_1, dies at rate μ_1, and converts one-way into type 2 at rate ν; type 2 replicates at rate λ_2 and dies at rate μ_2. In state (x, y) a single global catastrophe occurs at rate $\delta_1 x + \delta_2 y$ and sends the process to the absorbing cemetery (R, R). Write τ_c for the catastrophe time (with the usual convention $\tau_c = \infty$ on paths that never absorb).

Let $i \in \{1, 2\}$ label the initial type, and write $\mathbb{P}_i$, $\mathbb{E}_i$ for the law and expectation started from one individual of that type (so $i = 1$ means $(X_0, Y_0) = (1, 0)$ and $i = 2$ means $(0, 1)$).

Definition 1 (Ultimate release counts)**.** The ultimate type-1 and type-2 release counts are the random variables

$$(X_{\mathrm{rel}}, Y_{\mathrm{rel}}) \;=\; \begin{cases} (X_{\tau_c-},\, Y_{\tau_c-}) & \text{on } \{\tau_c < \infty\}, \\ (0,0) & \text{on } \{\tau_c = \infty\}. \end{cases} \tag{1}$$

Thus release records the left-limit population at the absorbing event when that event occurs, and is set to zero when no release event ever takes place—in particular when the intracellular population dies out by ordinary deaths before catastrophe.

Remark 1 (Generic regime versus boundaries)*.* In the generic regime $\delta_1 > 0$, $\delta_2 > 0$, the only way to have $\tau_c = \infty$ is ordinary extinction to $(0,0)$ before the catastrophe clock fires. Consequently the atom of $(X_{\mathrm{rel}}, Y_{\mathrm{rel}})$ at the origin is exactly the forever-avoidance probability, and coincides with clearance without release. On certain catastrophe-rate boundaries (for instance $\delta_1 = 0$), paths with $\tau_c = \infty$ need not be extinct; equation (1) still assigns $(0,0)$ on those paths, now meaning "no release event" rather than "empty burst". All formulae below remain valid on those boundaries; only the biological reading of the origin atom changes.

2 Killed generating function and catastrophe flux

Recall the killed (defective) generating functions

$$\mathcal{F}_1(x, y, t) = \mathbb{E}_{(1,0)}\left[x^{X_t} y^{Y_t} \mathbf{1}_{\{\tau_c > t\}}\right], \tag{2a}$$

$$\mathcal{F}_2(x, y, t) = \mathbb{E}_{(0,1)}\left[x^{X_t} y^{Y_t} \mathbf{1}_{\{\tau_c > t\}}\right], \tag{2b}$$

equivalently the occupation-time transforms

$$\mathcal{F}_i(x,y,t) = \mathbb{E}_i\left[x^{X_t} y^{Y_t} \exp\left\{-\int_0^t (\delta_1 X_s + \delta_2 Y_s)\,\mathrm{d}s\right\}\right].$$

Their coefficients and total masses are

$$p^{(i)}_{m,n}(t) := [x^m y^n]\mathcal{F}_i(x,y,t) = \mathbb{P}_i\{X_t = m,\, Y_t = n,\, \tau_c > t\}, \tag{3}$$

$$\sigma_i(t) := \mathcal{F}_i(1,1,t) = \begin{cases} S(t), & i = 1, \\ G(t), & i = 2, \end{cases} \tag{4}$$

so that $1 - \sigma_i(t) = \mathbb{P}_i\{\tau_c \le t\}$. Write

$$\sigma_i(\infty) := \lim_{t\to\infty} \sigma_i(t) = \begin{cases} h, & i = 1, \\ r_{2,-}, & i = 2, \end{cases} \tag{5}$$

for the forever-avoidance probabilities of the main theorem.

The catastrophe-flux generating function of the appendix is

$$\mathcal{J}_i(x,y,t) = \delta_1 x\, \partial_x \mathcal{F}_i(x,y,t) + \delta_2 y\, \partial_y \mathcal{F}_i(x,y,t). \tag{6}$$

Coefficientwise,

$$j^{(i)}_{m,n}(t) := [x^m y^n]\mathcal{J}_i(x,y,t) = (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t). \tag{7}$$

From state (m,n) the process is absorbed at rate $\delta_1 m + \delta_2 n$, so $j^{(i)}_{m,n}(t)$ is the joint density (in t) of the event

$$\{\tau_c \in \mathrm{d}t,\ X_{\tau_c-} = m,\ Y_{\tau_c-} = n\}.$$

In particular $j^{(i)}_{0,0}(t) \equiv 0$, since the empty state has catastrophe rate zero. Summing over counts recovers the marginal catastrophe-time densities

$$\mathcal{J}_1(1,1,t) = -S'(t), \qquad \mathcal{J}_2(1,1,t) = -G'(t). \tag{8}$$

3 Joint distribution of the release counts

3.1 Point masses

Integrating the flux density over all future times yields the law of the pre-catastrophe state on the event that catastrophe eventually occurs. Adding the forever-avoidance atom at the origin completes the law of $(X_{\mathrm{rel}}, Y_{\mathrm{rel}})$.

Proposition 1 (Joint law of ultimate release)**.** *For each initial type $i \in \{1,2\}$ and all integers $m,n \ge 0$,*

$$\mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n\} = \begin{cases} \sigma_i(\infty), & (m,n) = (0,0), \\ \displaystyle\int_0^\infty (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t)\,\mathrm{d}t, & m+n \ge 1. \end{cases} \tag{9}$$

Equivalently, for every $(m,n) \ne (0,0)$,

$$\mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n\} = \int_0^\infty j^{(i)}_{m,n}(t)\,\mathrm{d}t, \tag{10}$$

while the origin mass is

$$\mathbb{P}_i\{X_{\mathrm{rel}} = 0,\, Y_{\mathrm{rel}} = 0\} = \mathbb{P}_i\{\tau_c = \infty\} = \sigma_i(\infty). \tag{11}$$

Proof. On $\{\tau_c < \infty\}$ one has $(X_{\mathrm{rel}}, Y_{\mathrm{rel}}) = (X_{\tau_c-}, Y_{\tau_c-})$. The occupation measure of the first entrance to (R, R) from the pre-absorption state (m, n) is exactly the integrated flux equation (10). Since $j^{(i)}_{0,0} \equiv 0$, no catastrophe mass accumulates at the origin. On $\{\tau_c = \infty\}$ the definition assigns $(0, 0)$, which contributes the atom $\sigma_i(\infty)$. These two pieces partition the sample space, so the displayed probabilities sum to one:

$$\begin{aligned}\sum_{m,n\geq 0} \mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n\} &= \sigma_i(\infty) + \int_0^\infty \sum_{m,n} j^{(i)}_{m,n}(t)\,\mathrm{d}t \\ &= \sigma_i(\infty) + \int_0^\infty \mathcal{J}_i(1,1,t)\,\mathrm{d}t \\ &= \sigma_i(\infty) + \int_0^\infty (-\sigma_i'(t))\,\mathrm{d}t = \sigma_i(\infty) + (1 - \sigma_i(\infty)) = 1,\end{aligned}$$

where the last step uses $\sigma_i(0) = 1$ and $\sigma_i \downarrow \sigma_i(\infty)$. □

3.2 Joint probability generating function

Proposition 2 (Release PGF)**.** *The joint probability generating function of the ultimate release counts is*

$$\begin{aligned}\mathcal{P}^{\mathrm{rel}}_i(x,y) &:= \mathbb{E}_i\big[x^{X_{\mathrm{rel}}} y^{Y_{\mathrm{rel}}}\big] \\ &= \sigma_i(\infty) + \int_0^\infty \mathcal{J}_i(x,y,t)\,\mathrm{d}t \\ &= \sigma_i(\infty) + \int_0^\infty (\delta_1 x\,\partial_x + \delta_2 y\,\partial_y)\mathcal{F}_i(x,y,t)\,\mathrm{d}t.\end{aligned} \tag{12}$$

Consequently the joint masses of proposition 1 are recovered by extraction,

$$\mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n\} = [x^m y^n]\,\mathcal{P}^{\mathrm{rel}}_i(x,y). \tag{13}$$

Proof. Expand $\mathcal{J}_i$ by equation (7) and integrate term by term. The constant term of the integral vanishes, so the origin coefficient of $\mathcal{P}^{\mathrm{rel}}_i$ is exactly $\sigma_i(\infty)$; every other coefficient is the integrated flux of equation (10). □

Remark 2 (Finite horizon)*.* If one stops at a fixed observational horizon $T < \infty$ and records

$$(X^{(T)}_{\mathrm{rel}}, Y^{(T)}_{\mathrm{rel}}) = \begin{cases}(X_{\tau_c-}, Y_{\tau_c-}) & \text{on } \{\tau_c \leq T\},\\ (0,0) & \text{on } \{\tau_c > T\},\end{cases}$$

the same argument yields the truncated generating function

$$\mathcal{P}^{\mathrm{rel}}_{i,T}(x,y) = \sigma_i(T) + \int_0^T \mathcal{J}_i(x,y,t)\,\mathrm{d}t, \tag{14}$$

with $\mathcal{P}^{\mathrm{rel}}_i = \lim_{T\to\infty} \mathcal{P}^{\mathrm{rel}}_{i,T}$ pointwise for $|x| \leq 1$, $|y| \leq 1$.

4 Conditional laws and decomposition

The appendix records the distribution of the pre-catastrophe state *conditional on catastrophe*. That object is the catastrophe part of equation (12), renormalised.

Proposition 3 (Catastrophe-conditional release)**.** *Assume $\sigma_i(\infty) < 1$. The law of $(X_{\mathrm{rel}}, Y_{\mathrm{rel}})$ conditional on eventual catastrophe is supported on $\{(m,n) : m+n \geq 1\}$ and has probability generating function*

$$\mathcal{R}_i^{\mathrm{cat}}(x,y) := \mathbb{E}_i\big[x^{X_{\mathrm{rel}}} y^{Y_{\mathrm{rel}}} \mid \tau_c < \infty\big] = \frac{\int_0^\infty \mathcal{J}_i(x,y,t)\,\mathrm{d}t}{1-\sigma_i(\infty)}. \tag{15}$$

Equivalently, for $m+n \geq 1$,

$$\mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n \mid \tau_c < \infty\} = \frac{\int_0^\infty j_{m,n}^{(i)}(t)\,\mathrm{d}t}{1-\sigma_i(\infty)}. \tag{16}$$

The unconditional release PGF therefore decomposes as the two-point mixture

$$\mathcal{P}_i^{\mathrm{rel}}(x,y) = \sigma_i(\infty)\cdot 1 + \big(1-\sigma_i(\infty)\big)\,\mathcal{R}_i^{\mathrm{cat}}(x,y), \tag{17}$$

i.e. an atom of mass $\sigma_i(\infty)$ at $(0,0)$, and with complementary mass the catastrophe-conditional release law of the appendix (the $T\to\infty$ limit of $\mathcal{R}_{i,T}^{\mathrm{cat}}$).

Proof. On $\{\tau_c < \infty\}$ one has $(X_{\mathrm{rel}}, Y_{\mathrm{rel}}) = (X_{\tau_c-}, Y_{\tau_c-})$, so equation (15) is the integrated and renormalised flux already identified in the appendix. Equation (17) is then immediate from equation (12). $\square$

Time-resolved versions are likewise available from the appendix: conditional on $\tau_c = t$ in the density sense,

$$\mathbb{E}_i\big[x^{X_{\tau_c-}} y^{Y_{\tau_c-}} \mid \tau_c = t\big] = \frac{\mathcal{J}_i(x,y,t)}{\mathcal{J}_i(1,1,t)}, \tag{18}$$

and therefore

$$\mathbb{P}_i\{X_{\tau_c-} = m,\, Y_{\tau_c-} = n \mid \tau_c = t\} = \frac{j_{m,n}^{(i)}(t)}{\mathcal{J}_i(1,1,t)}. \tag{19}$$

Integrating equation (18) against the catastrophe-time density $\mathcal{J}_i(1,1,t) = -\sigma_i'(t)$ recovers the numerator of equation (15).

5 Marginal distributions

The marginal generating functions are the obvious specialisations of $\mathcal{P}_i^{\mathrm{rel}}$.

Corollary 1 (Marginals of release)**.** *For each initial type i,*

$$\mathbb{E}_i\big[x^{X_{\mathrm{rel}}}\big] = \mathcal{P}_i^{\mathrm{rel}}(x,1) = \sigma_i(\infty) + \int_0^\infty \delta_1 x\,\partial_x \mathcal{F}_i(x,1,t)\,\mathrm{d}t + \int_0^\infty \delta_2\,\partial_y \mathcal{F}_i(x,1,t)\,\mathrm{d}t, \tag{20}$$

$$\mathbb{E}_i\big[y^{Y_{\mathrm{rel}}}\big] = \mathcal{P}_i^{\mathrm{rel}}(1,y) = \sigma_i(\infty) + \int_0^\infty \delta_1\,\partial_x \mathcal{F}_i(1,y,t)\,\mathrm{d}t + \int_0^\infty \delta_2 y\,\partial_y \mathcal{F}_i(1,y,t)\,\mathrm{d}t. \tag{21}$$

The corresponding point masses are

$$\mathbb{P}_i\{X_{\mathrm{rel}} = m\} = [x^m]\,\mathcal{P}_i^{\mathrm{rel}}(x,1), \tag{22}$$

$$\mathbb{P}_i\{Y_{\mathrm{rel}} = n\} = [y^n]\,\mathcal{P}_i^{\mathrm{rel}}(1,y). \tag{23}$$

In particular both marginals carry the common origin atom $\sigma_i(\infty)$, since $(X_{\text{rel}}, Y_{\text{rel}}) = (0,0)$ on $\{\tau_c = \infty\}$. Explicitly, for $m \geq 1$,

$$\mathbb{P}_i\{X_{\text{rel}} = m\} = \int_0^\infty \sum_{n=0}^{\infty} (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t)\, \mathrm{d}t, \tag{24}$$

and for $n \geq 1$,

$$\mathbb{P}_i\{Y_{\text{rel}} = n\} = \int_0^\infty \sum_{m=0}^{\infty} (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t)\, \mathrm{d}t, \tag{25}$$

while

$$\mathbb{P}_i\{X_{\text{rel}} = 0\} = \sigma_i(\infty) + \int_0^\infty \sum_{n=1}^{\infty} \delta_2 n\, p^{(i)}_{0,n}(t)\, \mathrm{d}t, \tag{26}$$

and likewise

$$\mathbb{P}_i\{Y_{\text{rel}} = 0\} = \sigma_i(\infty) + \int_0^\infty \sum_{m=1}^{\infty} \delta_1 m\, p^{(i)}_{m,0}(t)\, \mathrm{d}t. \tag{27}$$

(The second summands are the probabilities of catastrophe from a pure type-2 or pure type-1 state, respectively; they vanish if the corresponding catastrophe rate is zero.)

Remark 3 (Joint origin versus marginal origins)*.* The event $\{X_{\text{rel}} = 0\}$ is strictly larger than $\{X_{\text{rel}} = Y_{\text{rel}} = 0\}$ whenever catastrophe can occur from a state with $X = 0$ and $Y \geq 1$. Thus $\mathbb{P}_i\{X_{\text{rel}} = 0\} \geq \sigma_i(\infty)$, with equality if and only if pure type-2 states cannot trigger (in particular if $\delta_2 = 0$). The same remark applies to Y_{rel} with the roles of the types reversed.

6 Means and low-order moments

Differentiating the release PGF at $(1,1)$ yields moments of the ultimate release. Because $\mathcal{P}_i^{\text{rel}}$ has an atom at the origin, these moments are *unconditional* means of released load (including zeros on forever-avoidance paths). Conditional means given catastrophe are obtained by dividing by $1 - \sigma_i(\infty)$.

Corollary 2 (Mean released load)**.** *Whenever the derivatives exist in a neighbourhood of* $(1,1)$*,*

$$\mathbb{E}_i[X_{\text{rel}}] = \partial_x \mathcal{P}_i^{\text{rel}}(1,1) = \int_0^\infty \Big(\delta_1\, \partial_x(x\, \partial_x \mathcal{F}_i) + \delta_2\, \partial_x\big(y\, \partial_y \mathcal{F}_i\big)\Big)\Big|_{(x,y)=(1,1)} \mathrm{d}t, \tag{28}$$

$$\mathbb{E}_i[Y_{\text{rel}}] = \partial_y \mathcal{P}_i^{\text{rel}}(1,1) = \int_0^\infty \Big(\delta_1\, \partial_y\big(x\, \partial_x \mathcal{F}_i\big) + \delta_2\, \partial_y\big(y\, \partial_y \mathcal{F}_i\big)\Big)\Big|_{(x,y)=(1,1)} \mathrm{d}t. \tag{29}$$

Equivalently, in coefficient form,

$$\mathbb{E}_i[X_{\text{rel}}] = \int_0^\infty \sum_{m,n \geq 0} m\, (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t)\, \mathrm{d}t, \tag{30}$$

$$\mathbb{E}_i[Y_{\text{rel}}] = \int_0^\infty \sum_{m,n \geq 0} n\, (\delta_1 m + \delta_2 n)\, p^{(i)}_{m,n}(t)\, \mathrm{d}t. \tag{31}$$

The conditional means given eventual catastrophe are

$$\mathbb{E}_i[X_{\text{rel}} \mid \tau_c < \infty] = \frac{\mathbb{E}_i[X_{\text{rel}}]}{1 - \sigma_i(\infty)}, \qquad \mathbb{E}_i[Y_{\text{rel}} \mid \tau_c < \infty] = \frac{\mathbb{E}_i[Y_{\text{rel}}]}{1 - \sigma_i(\infty)}. \tag{32}$$

(The origin atom does not contribute to either mean.)

7 Evaluation from the closed killed PGF

The formulae above are distributional identities: they do not require a closed form for $\mathcal{F}_i$. When the hypergeometric solution of the appendix is available, they become explicit integral transforms of that solution.

In the generic regime of the main theorem, after the type-1 time scaling $\hat{t} = \lambda_1 t$,

$$\widehat{\mathcal{F}}_2(x, y, \hat{t}) = r_{2,-} - \alpha_2 \frac{z_2(\hat{t}; y)}{1 - z_2(\hat{t}; y)}, \tag{33}$$

$$\widehat{\mathcal{F}}_1(x, y, \hat{t}) = h - \hat{q}_2 z \frac{K_1 W_1'(z) + K_2 W_2'(z)}{K_1 W_1(z) + K_2 W_2(z)}, \quad z = z_2(\hat{t}; y), \tag{34}$$

with $z_{2,0}(y) = (y - r_{2,-})/(y - r_{2,+})$, $z_2(\hat{t}; y) = z_{2,0}(y)\, e^{\hat{q}_2 \hat{t}}$, and Wronskian coefficients (K_1, K_2) built from the initial logarithmic derivative $L(x, y) = (h - x)/(\hat{q}_2 z_{2,0}(y))$ as in the appendix. Substituting equations (33) and (34) into equation (6) and integrating in physical time (or in $\hat{t}$, with the Jacobian $1/\lambda_1$) yields $\mathcal{P}_i^{\mathrm{rel}}$ as an explicit improper integral of elementary and hypergeometric functions of (x, y). Coefficients may then be read by a bivariate Cauchy integral on a product of small circles about the origin, or by series expansion of $\mathcal{F}_i$ before the time integral when a power-series route is preferred.

For numerical work at fixed (x, y), one may equally integrate the killed backward system of the appendix for $\mathcal{F}_i(x, y, \cdot)$ and accumulate the flux integrand of equation (12) in parallel, without extracting coefficients first. The joint masses equation (9) are the special case of integrating the scalar ODEs for each coefficient $p_{m,n}^{(i)}(t)$ against the weight $\delta_1 m + \delta_2 n$.

8 Summary

The ultimate release pair $(X_{\mathrm{rel}}, Y_{\mathrm{rel}})$ is completely determined by the killed generating function through its catastrophe flux:

$$\boxed{\begin{aligned} \mathcal{P}_i^{\mathrm{rel}}(x, y) &= \mathbb{E}_i\big[x^{X_{\mathrm{rel}}} y^{Y_{\mathrm{rel}}}\big] = \sigma_i(\infty) + \int_0^\infty \mathcal{J}_i(x, y, t)\,\mathrm{d}t, \\ \mathbb{P}_i\{X_{\mathrm{rel}} = m,\, Y_{\mathrm{rel}} = n\} &= [x^m y^n]\, \mathcal{P}_i^{\mathrm{rel}}(x, y) \\ &= \begin{cases} \sigma_i(\infty), & (m, n) = (0, 0), \\ \displaystyle\int_0^\infty (\delta_1 m + \delta_2 n)\, p_{m,n}^{(i)}(t)\,\mathrm{d}t, & m + n \geq 1. \end{cases} \end{aligned}} \tag{35}$$

The catastrophe-conditional law of the appendix is the same object with the origin atom removed and the remaining mass renormalised by $1 - \sigma_i(\infty)$. Marginals and moments follow by specialising or differentiating $\mathcal{P}_i^{\mathrm{rel}}$.